Hydra Hub – User Manual

1. Introduction

Hydra Hub is a management and operation platform for **Hydra Heads** on Cardano. It allows users to create Projects, configure Heads, manage Nodes/Participants, and monitor Hydra connection status in real time.

This document provides an **end-to-end** guide for users, from account registration to running a Hydra Head and successfully establishing a connection.

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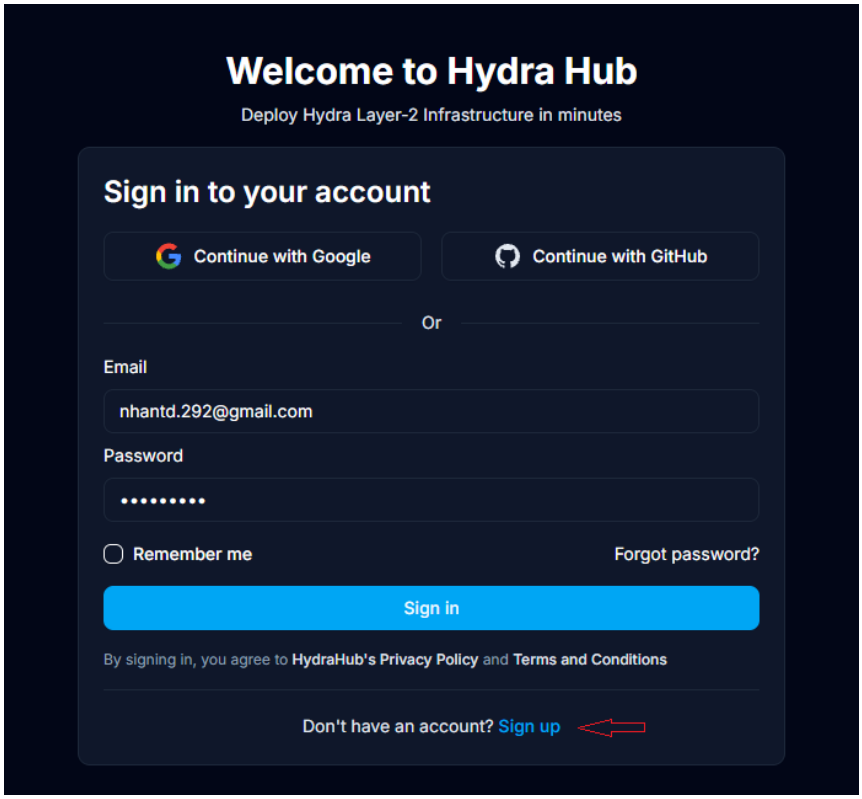
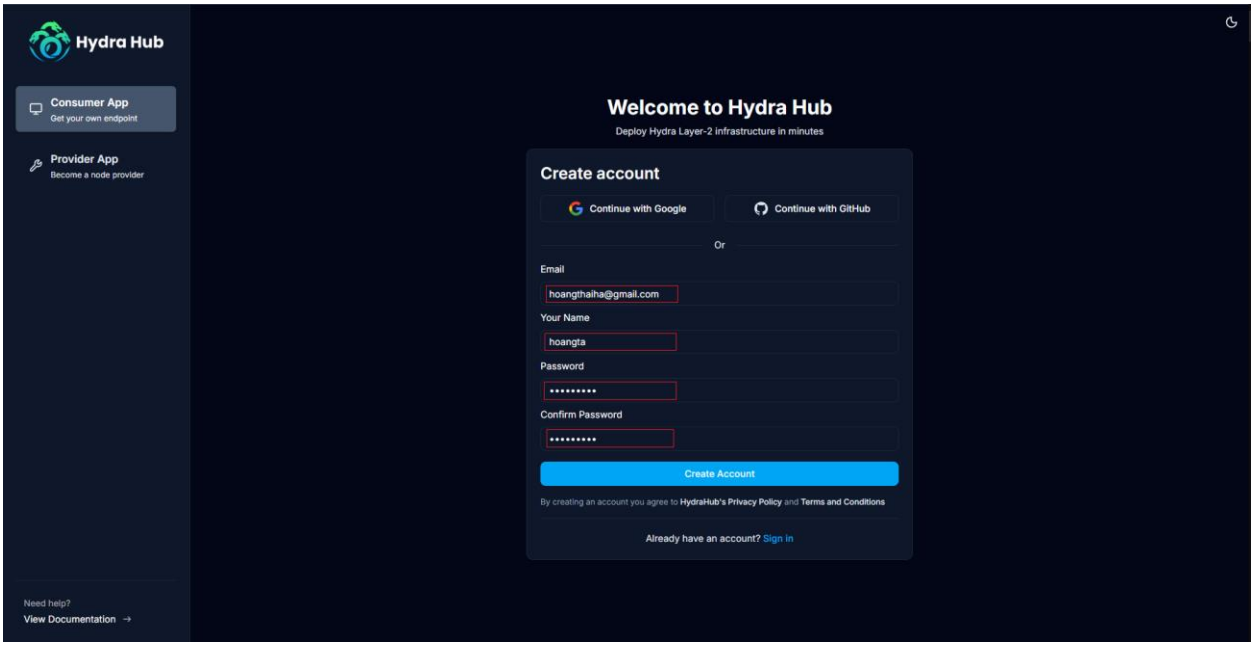
2. Sign Up & Sign In

2.1 Account Registration

- Access Hydra Hub at https://uat.hydrahub.io.vn/



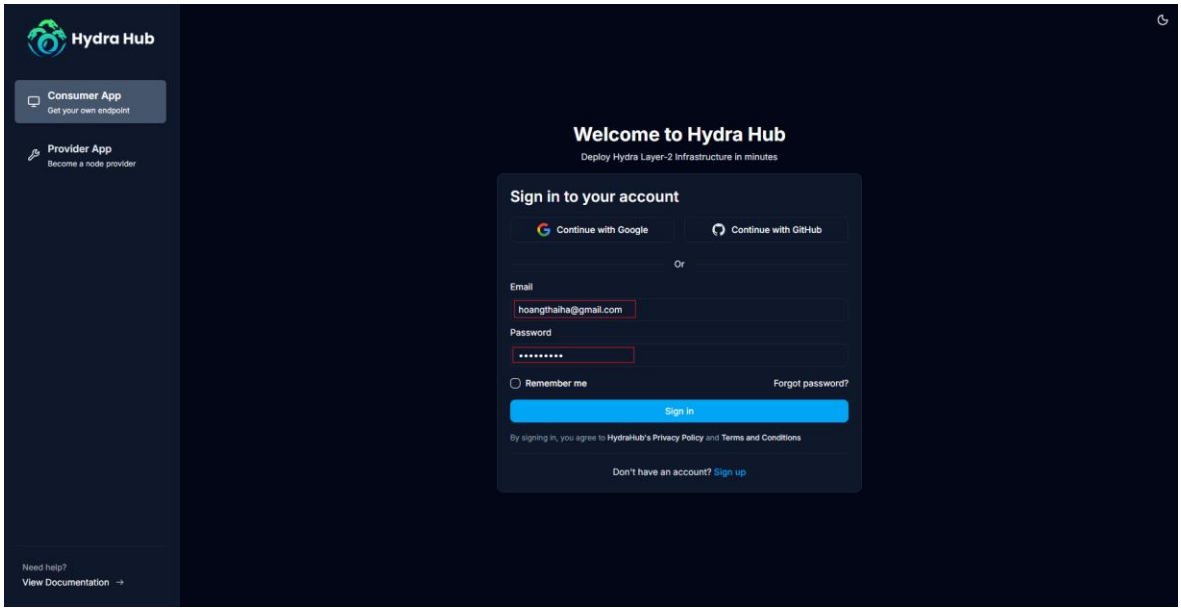
- Click **Sign up**



- Fill in the required account information and submit the request
- The account status will be **Pending** until approved by an Admin

2.2 Sign In

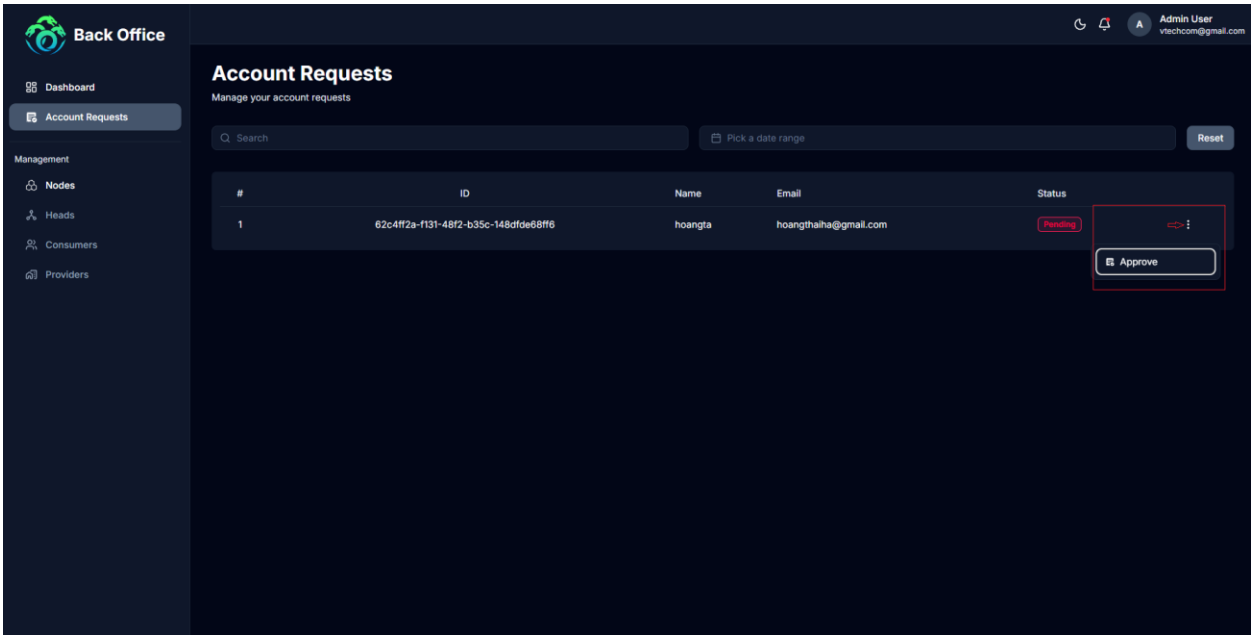
- After the account is approved by the Admin
- Sign in using your email and password



3. Admin Account Approval

This step is for Admin users only

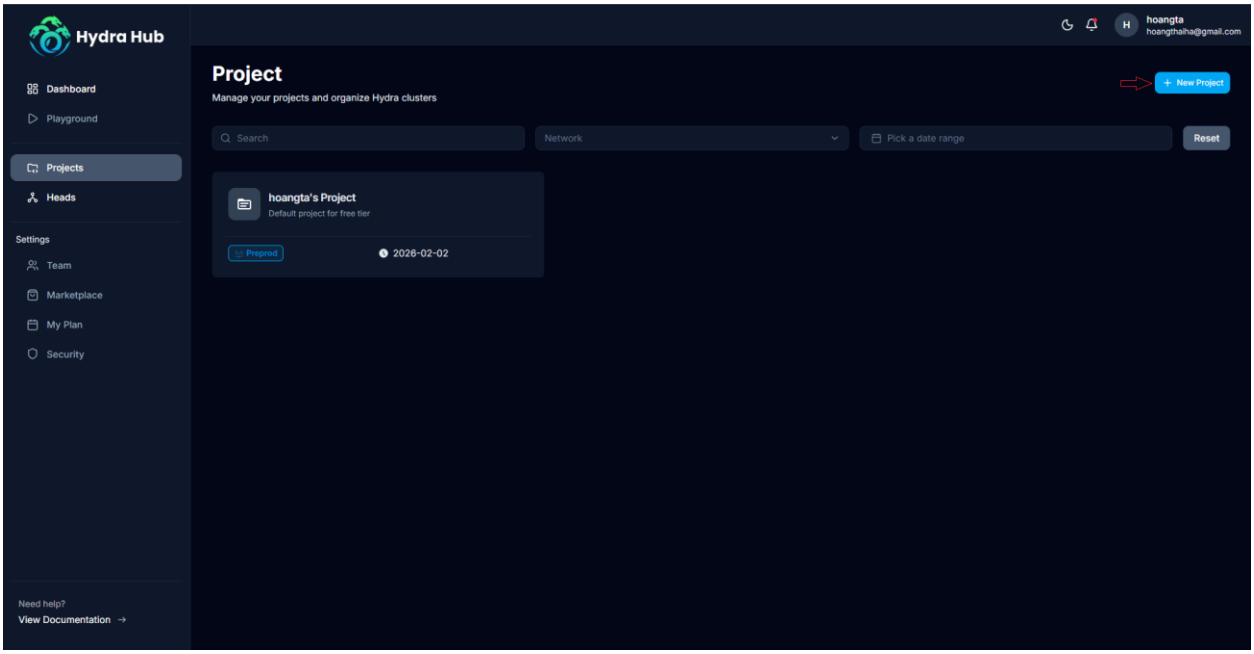
- Go to **Admin Dashboard** → **Account Requests**
- Review the list of pending accounts
- Click **Approve** to activate the account



4. Project Management

4.1 Create a Project

- Open the **Projects** menu, where you can view the list of projects you have created
- Click **New Project**
    - By default, a project is automatically created when you register an account
    - To create and manage multiple projects, you may need to upgrade your plan



- Each Project includes a **Project API Key**, which is used to authenticate requests

4.2 Project API Key

- The API Key must be included in the request header:

```
X-Api-Key: <proj_0618****54d064dea>
```

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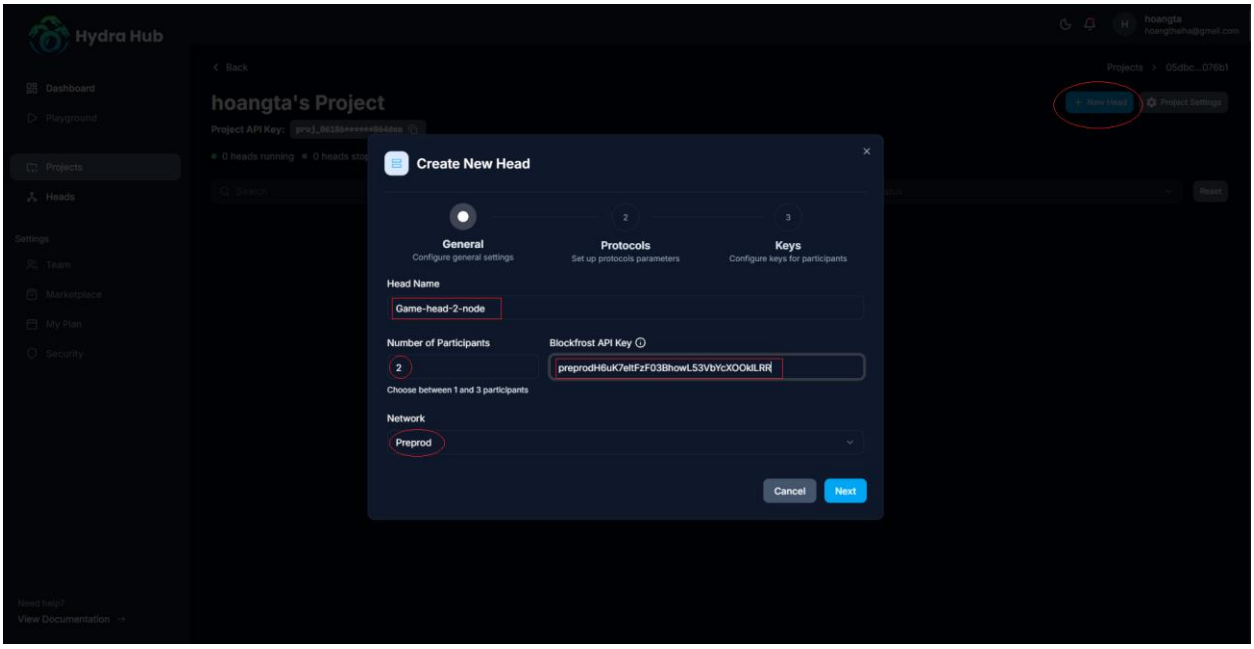
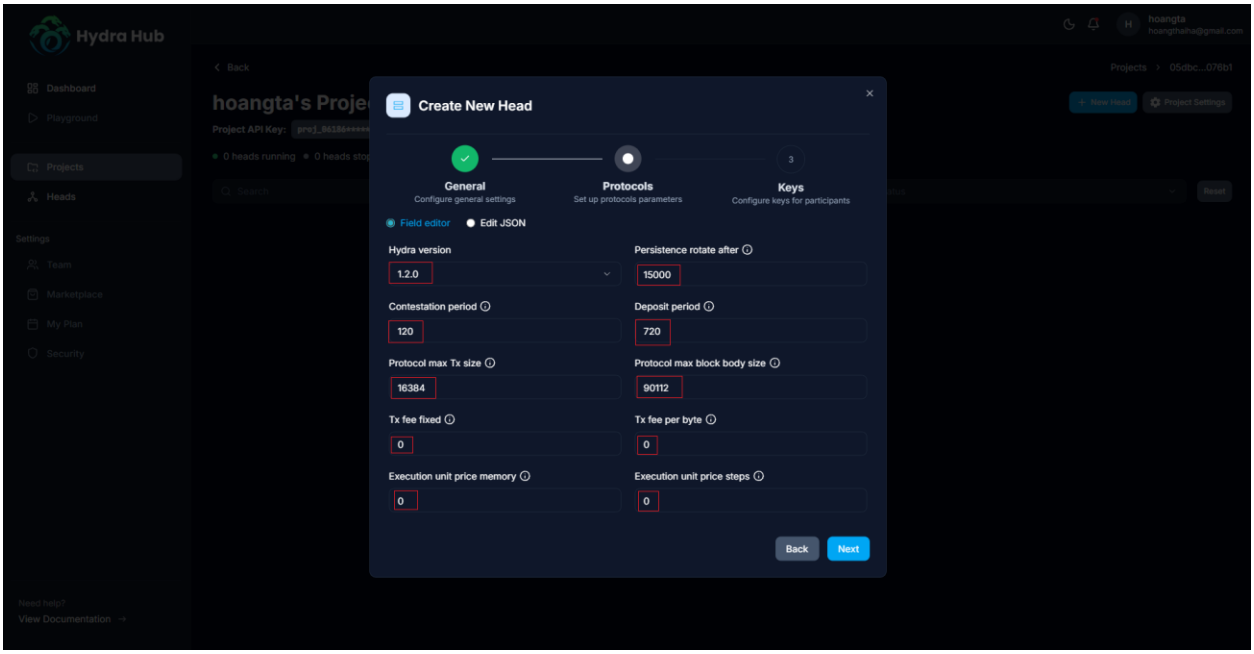
5. Hydra Head

5.1 Create a Head

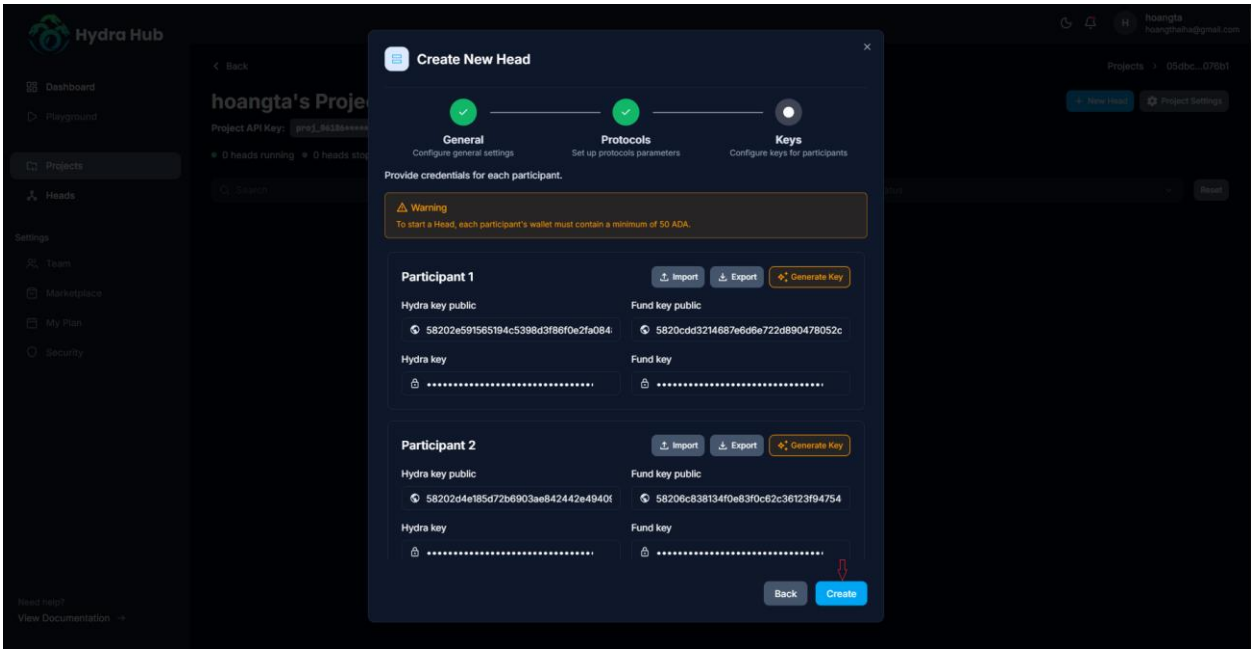
- Inside a Project, click **New Head**
- Configure the Head step by step:
    - **General**: Head name and environment (Preprod/Mainnet)



- **Protocols**: Hydra version, contestation period, deposit period



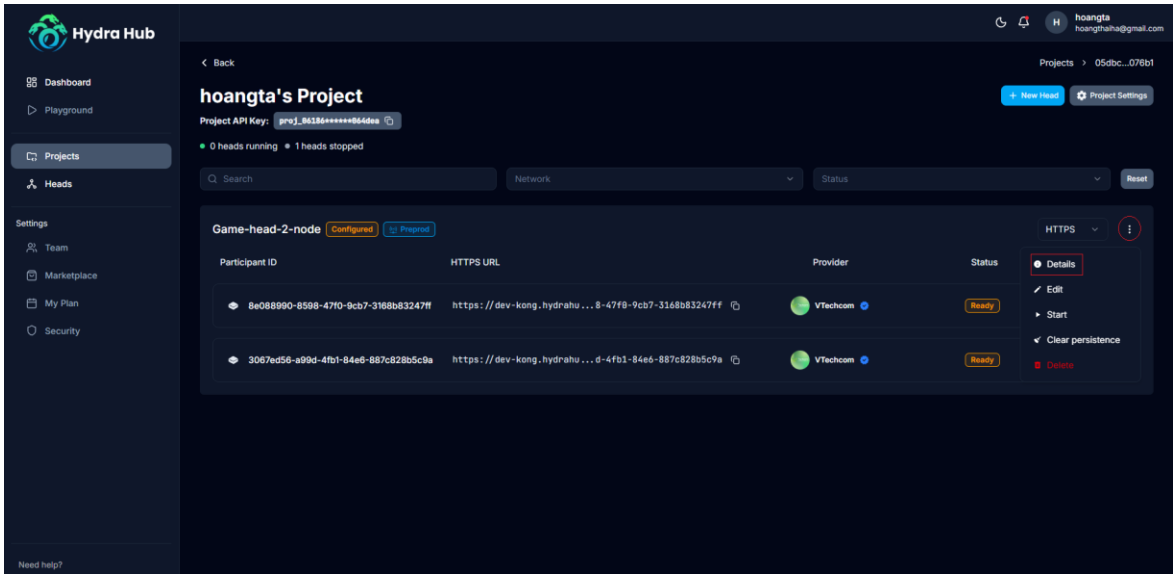
- **Keys**: Verification keys and persistence settings



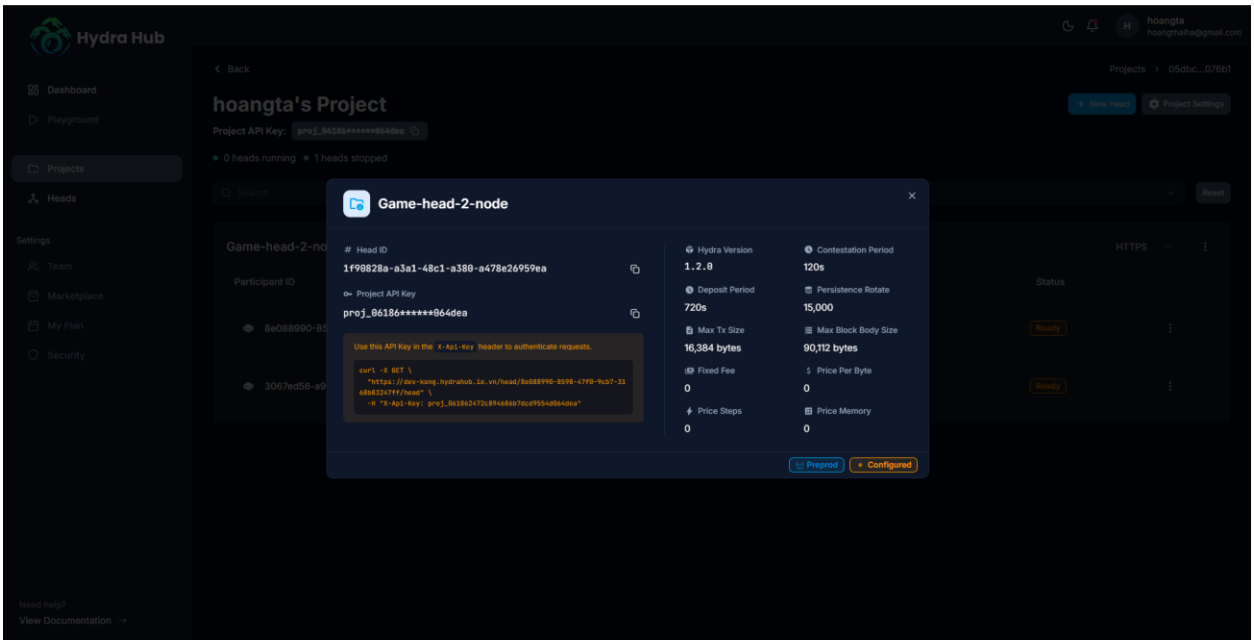
5.2 Head Status

- `Configured`: Configuration completed
- `Ready`: Ready to be started
- `Running`: Head is running

5.3 Available Actions



- **Details**: View Head details



- **Edit**: Edit the display name of the Head
- **Start / Stop**: Start or pause the Head
- **Restart**: Restart the Head
- **Delete**: Delete the Head

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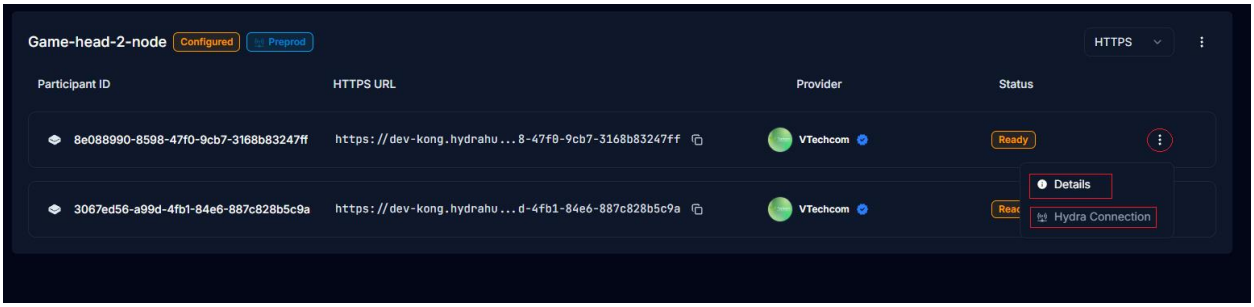
6. Node / Participant Management

- Each Head consists of multiple Participants
- Displayed information includes:



- Participant ID
- HTTPS URL
- Provider
- Status

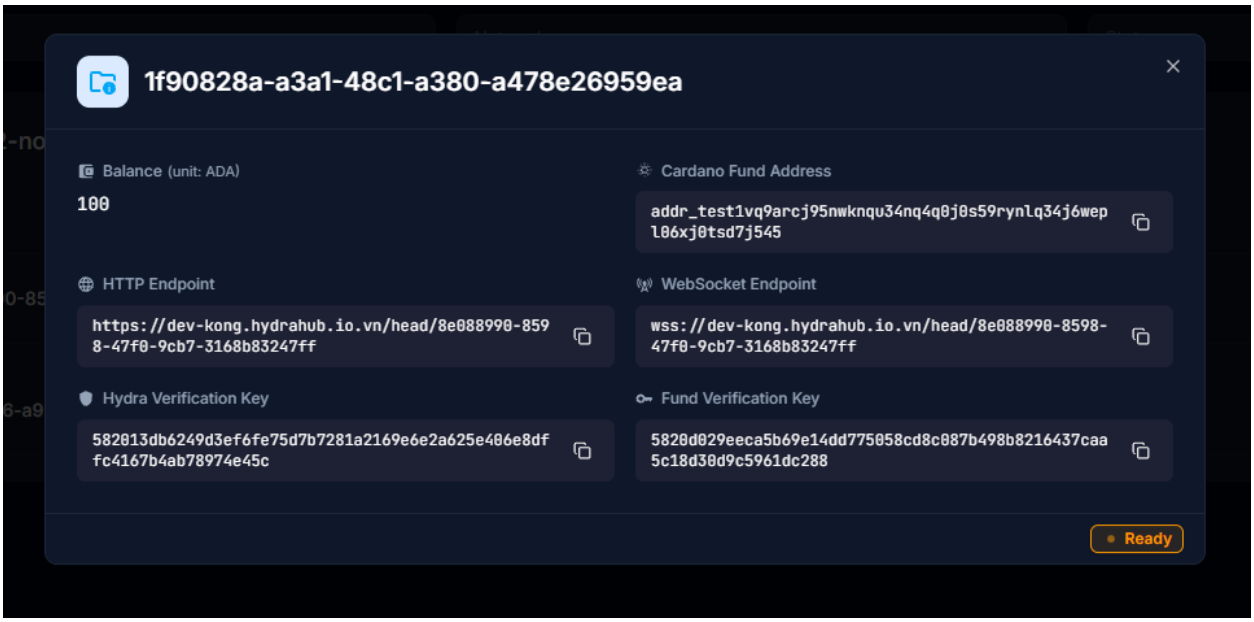
6.1 Available Actions

- **Details**: View Node details
- **Hydra Connection**: Connect to the Hydra node

6.2 Node Details

The Node Details screen displays:

- Balance (ADA)
- Cardano Fund Address
- HTTP Endpoint
- WebSocket Endpoint
- Hydra Verification Key
- Fund Verification Key



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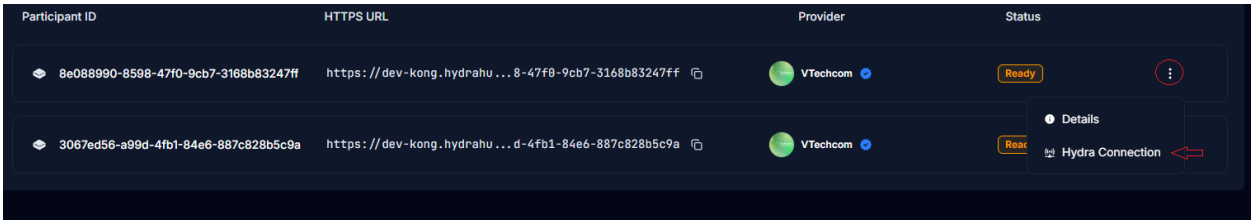
7. Hydra Connection

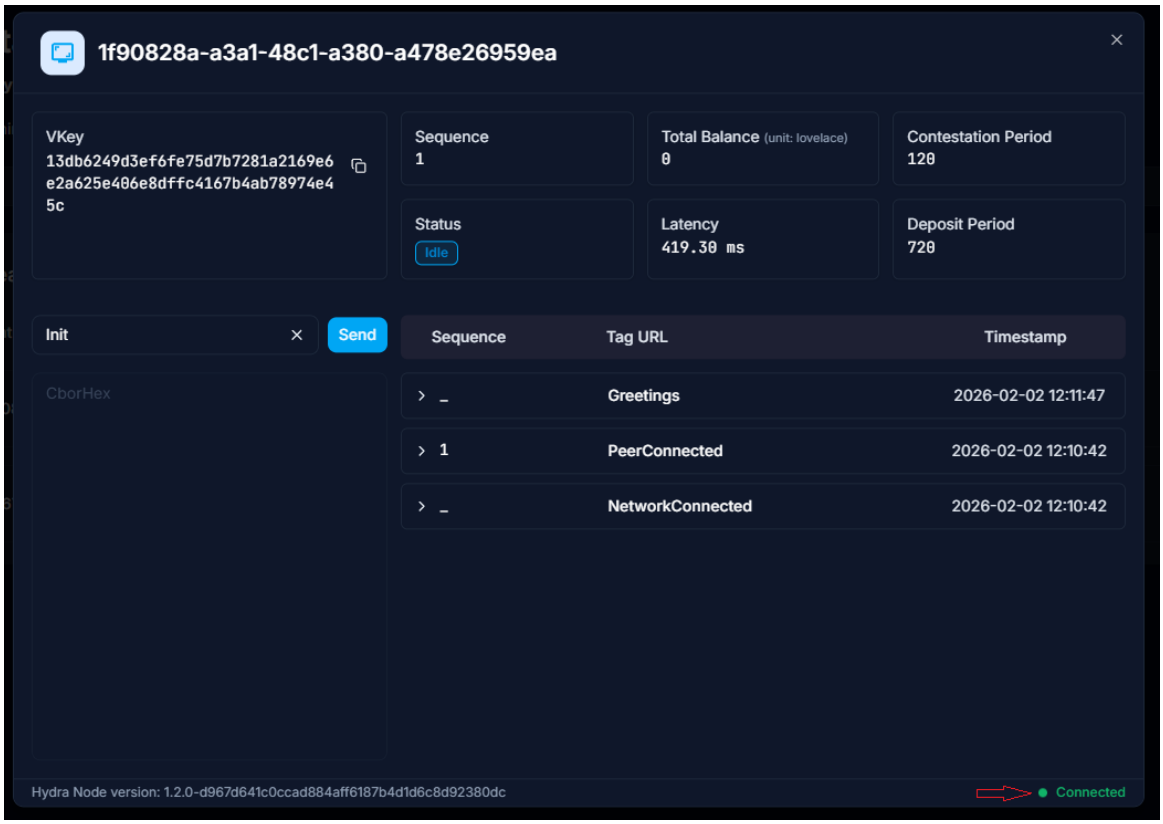
7.1 Connect

- Select **Hydra Connection**
- The system opens a real-time WebSocket connection



7.2 Status

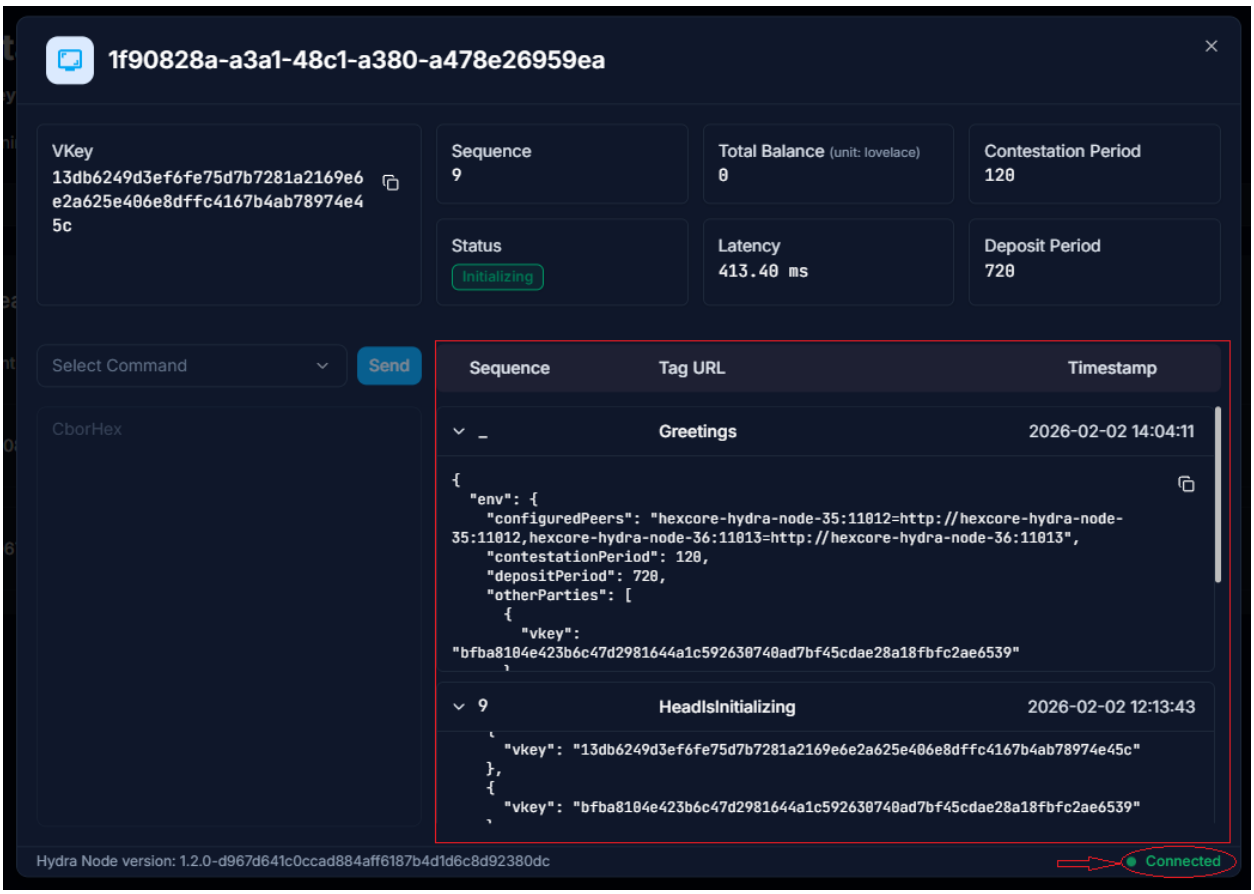
- Idle
- Initializing
- Connected



7.3 Event Logs

Common events include:

- `Greetings`
- `NetworkConnected`
- `PeerConnected`
- `HeadIsInitializing`



When the bottom-right corner shows **Connected (green)**, the Head is running successfully.

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8. Notes & Best Practices

- Always keep your API Key secure
- Only start a Head when all Participants are in the `Ready` state
- Use Preprod for testing before deploying to Mainnet

9. Support

- Refer to the **Documentation** available in Hydra Hub
- Contact the operations team if you encounter any issues